

Introduction to Natural Language Processing

Lecture Notes — EduSphere LMS

What is NLP?

Natural Language Processing (NLP) is a subfield of artificial intelligence that focuses on enabling computers to understand, interpret, and generate human language. It sits at the intersection of computer science, linguistics, and machine learning.

Why NLP Matters

- Powers virtual assistants like Siri and Alexa
- Enables machine translation between languages
- Drives sentiment analysis for reviews and social media
- Supports chatbots and customer support automation
- Forms the foundation for modern Large Language Models (LLMs)

Core NLP Tasks

- **Tokenization** — splitting text into words or subwords
- **Part-of-Speech Tagging** — identifying nouns, verbs, adjectives, etc.
- **Named Entity Recognition** — detecting names, places, dates
- **Sentiment Analysis** — determining emotional tone of text
- **Machine Translation** — converting text between languages

Basic NLP Pipeline

1. Text Collection → 2. Text Preprocessing (cleaning, lowercasing, removing stopwords) → 3. Tokenization → 4. Feature Extraction (e.g. embeddings) → 5. Model Training / Inference → 6. Output (classification, generation, translation, etc.)

Next Lesson: We'll explore how deep learning models process tokenized text to perform sentiment analysis and other classification tasks.